Labor Force Non-Participation in a Search and Matching Model

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Motivation

- Recent evidence: labor force participation fluctuates over the business cycle
- Evidence that discouraged workers leave labor force, but little theoretical motivation for this phenomenon
- Why should we care?
  - Want to explain a feature of data
  - Model of non-participation important for evaluating policy
- This project: fluctuating human capital
  $\Longrightarrow$ some workers cannot be matched (discouraged)
- Declining human capital during unemployment $\Longrightarrow$ prolonged unemployment leads to non-participation
- Introduce aggregate fluctuations into standard matching model with human capital and heterogeneous firms

The Literature: Two Strands

- Various papers by Sahin and coauthors
  - Flows from nonparticipation to unemployment *increase* during recessions!
  - Decomposition of data: marginally attached workers join the labor market
  - Suggestions: search intensity, indivisible labor, insurance role of non-participating members of households...
  - But also: composition of non-participation includes more prime-age males during recessions $\implies$ movement from labor force outwards
- Kroft et al. (2016, 2019)
  - All types of workers more likely to be non-participating after long unemployment spells $\iff$ duration dependence in unemployment and non-participation

Contribution to Literature

- *Model* which focuses specifically on discouraged workers
- Explicit definition of discouragement
- Includes duration dependence in human capital

Model Parameters and Variables: Toy Example

- State variables:
  - Human capital h
  - $h \in \{\underline{h}, \overline{h}\}$ such that workers less likely to transition to high state $\overline{h}$ while unemployed
- Transition to unemployment with probability λ
- Hired at rate $p = q * \chi(V^E > V^U) * \chi(V^F > 0)$
- Fixed home production b

Value Functions and other details

- Unemployment:
  $J^U(h) = b + \beta \mathbb{P}(match)\, \mathbb{E}_h\left[J^E\right] + \beta\left(1 - \mathbb{P}(match)\right) \mathbb{E}_h\left[J^U\right]$
- Employment: $J^E(h) = w(h) + \beta\lambda \mathbb{E}_h\left[J^U\right] + \beta\left(1-\lambda\right) \mathbb{E}_h\left[J^E\right]$
- Filled Job: $J^F = f(h) - w(h) + \beta\left(1-\lambda\right) \mathbb{E}_h\left[J^E\right]$
- Vacancy: $J^V(h) \equiv 0$
- Wage w determined by Nash Bargaining
- FOC: $J^F(h) = \frac{1-\alpha}{\alpha}\left(J^E(h) - J^U(h)\right)$

Detailed Explanation of the Mechanism

- Match probability: $\mathbb{P}(match) = \mathbb{P}(meeting) \times \mathbb{P}(offer) \times \mathbb{P}(accept)$
- $q \times \mathbb{P}\left(J^F > J^V\right) \times \mathbb{P}\left(J^E > J^U\right)$
- Non-retired NILF $\iff$ match probability $= 0$
- Conditions based on productivity of match $f(A, z, h)$
- This implies reservation productivities z^w, h^f
- $\mathbb{P}(match) = q \times \mathbb{P}\left(h \geq h^f\right) \times \mathbb{P}\left(z \geq z^w\right)$
- Note that this mechanism exists in every search model! But often leads to degenerate solutions
- E.g.: this idea shows up as a regularity condition with homogeneous workers and firms (so not everyone is unemployed)

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Solution Procedure: Two worker types, one firm

- Compute value functions for firm, workers
- Solve Nash Bargaining problem to get wages for each type (nonlinearly solve 2×2 system in this example)
- Check if hiring probability converges; if not, update
- Find stationary distribution

Results: Changing Benefits

b	w_h	w_l	q_h	q_l	E_h	E_l	U_h	U_l
1.0	2.78	1.56	0.45	0.45	0.27	0.48	0.08	0.17
1.9	2.87	1.69	0.45	0.45	0.27	0.48	0.08	0.17
2.0	3.38	0.62	0.45	0	0.12	0.13	0.08	0
4.0	2.34	1.29	0	0	0	0	0.14	0.86

b	E	U	N
1.0	0.75	0.25	0
1.9	0.75	0.25	0
2.0	0.25	0.08	0.67
4.0	0	0	1

Full Model: State Variables

- Human capital through on-the-job learning
  - Human capital increases by η during every period of employment
  - Human capital decreases by δ during every period of unemployment
- Firm productivity z_i from fixed distribution $F(z)$.
  - Could be extended to endogenous firm distribution, though not common in the literature.
  - Enters twice into state space: current and previous offer
- Regularizing factor
  - With only on-the-job learning, nonparticipation is degenerate solution!
  - Aggregate shocks
  - Non-decreasing human capital during unemployment

Full Model: Calibration

- Many parameters are standard
- Difficult thing: how to calibrate human capital?
- Most papers: use administrative data
- Other sorts of human capital: use MEPS, SIPP, or CPS data to calibrate health
- Firm productivity distribution calibrated to firm sizes

Possible Experiments with Full Framework

- Policy experiments: simulate effects of policy reforms on stochastic steady state
  - Minimum wage: simulate with $w = \max\{\underline{w}, w_{NB}\}$
  - Unemployment benefits: simulate with new b' and approximated tax system $\tau(w)$
  - Job guarantee: simulate with b, $\tau(w)$, new rate of human capital depreciation δ'
- Extensions to model
  - Here: fixed q. Natural extension: $q(\theta)$ where θ is market tightness.
  - Here: fixed firm productivity distribution $F(z)$. Natural extension: simulate firm distribution as studied in class.
  - Here: workers are hand-to-mouth. Natural extension: allow for asset accumulation, effects on wages. (+ risk aversion)

Conclusion

- Idea: Come up with a model which can explain discouraged workers
- Model: Basic search model with duration dependence
- Why? Noticed in empirical literature + previously unexplained (to my knowledge)
- Proof of concept: model can deliver non-participation from low-productivity workers
- Larger model: include on-the-job search + learning, firm size distribution, endogenous hiring, aggregate shocks

The End

Thank you!

Appendix: Calibration

Calibration	Value
β: impatience	0.95
Transition to Bad State	0.30
Transition to Good State (E)	0.20
Transition to Good State (U)	0.05
λ: firing rate	0.145
q: base hiring rate	0.45
α: Bargaining power	0.72
Production function	$f(h) = h^{0.7}$